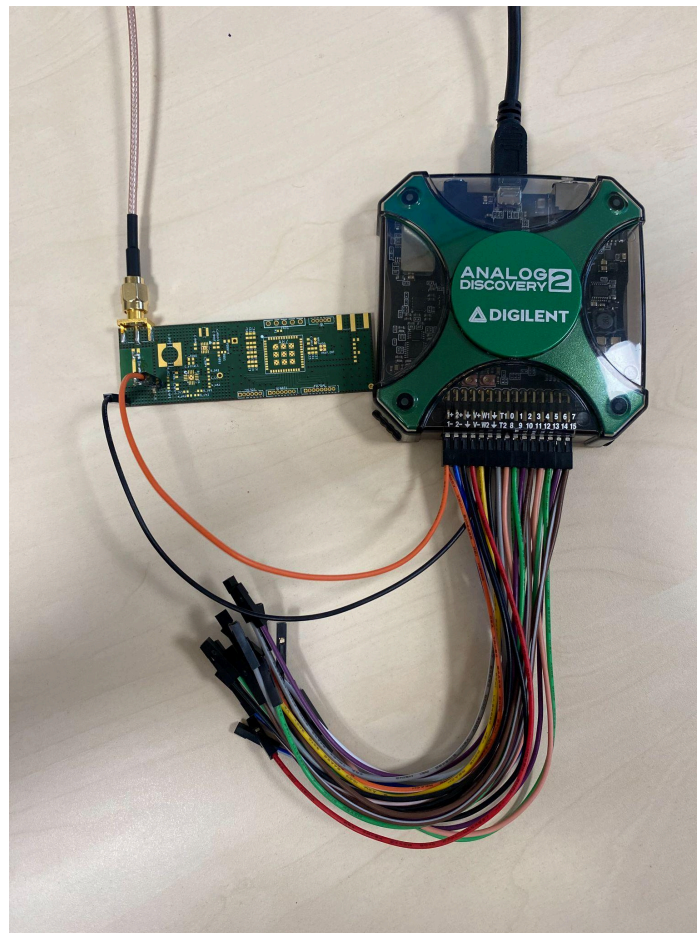


Energy for connected objects



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Introduction

This lab report explores methods for powering connected devices by harvesting energy from electromagnetic fields and transmitting power wirelessly. We investigated the power requirements of a small red light, tested approaches for direct energy usage and storage, and experimented with various technologies for energy management and conversion. The goal is to develop self-powered devices, enhancing charging efficiency, sensing capabilities, and secure data transmission.

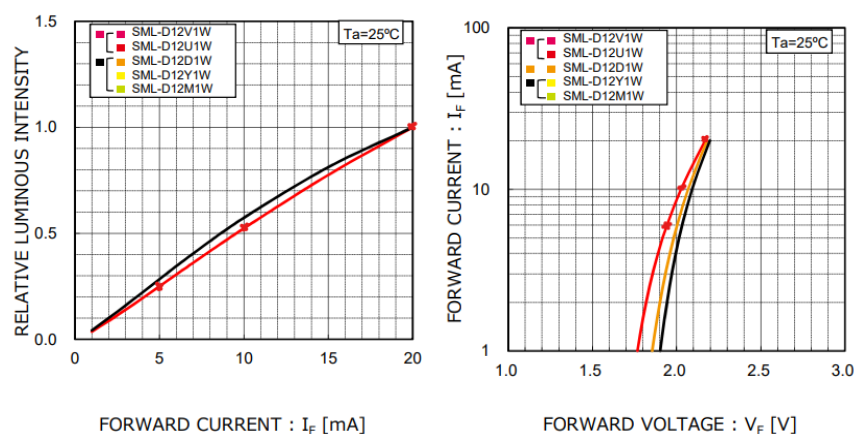
I. Study of the load and design

1- The DC power required by the LED SML-D12U1WT8 to work in its :

$$\text{Nominal use : } P = U.I = 2.2V * 20mA = 44mW$$

$$50\% \text{ of use : } P = U.I = 2V * 10mA = 20mW$$

$$25\% \text{ of use : } P = U.I = 1.9V * 5mA = 9,5mW$$



2- The energy required to light on the LED during 1s :

$$\text{Nominal (100\%): } E = P * t = 44mW * 1s = 44mJ$$

$$50\% : E = P * t = 20mW * 1s = 20mJ$$

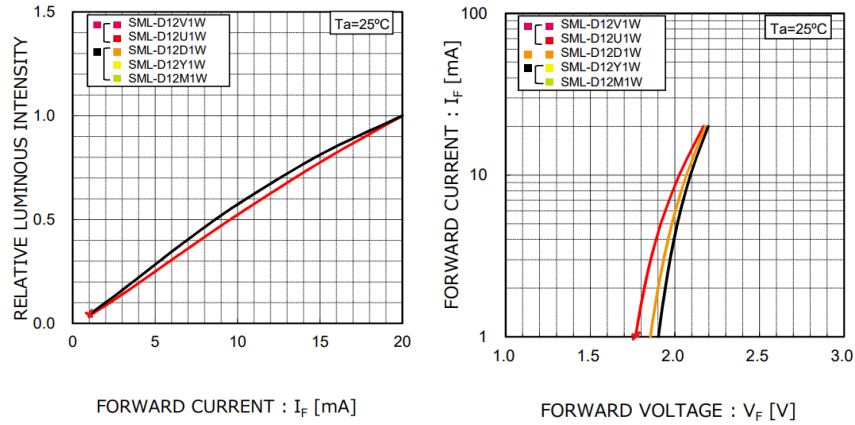
$$25\% : E = P * t = 9.5mW * 1s = 9.5mJ$$

3- The power provided to the LED to work :

$$I_{\min} = 1mA$$

$$V_{\min} = 1.75V$$

$$P_{\min} = 1.75V * 1mA = 1.75mW$$



The luminosity of the LED is less than 5% for minimum power.

4- Configurations of capacitance and activation and deactivation voltage thresholds that can be employed and the minimum input DC power required to work in the worst case :

We want to store the minimum unused energy. For this, we have to choose a low capacitor/voltage:

The power management unit (PMU) associated with a DC-DC converter requires power within specific voltage ranges to operate properly (between 2.2 V and 5.25 V).

The available supercapacitors come in different capacitance values: 100 μ F, 220 μ F, 1.5 mF, 2.2 mF, 6.8 mF, 10 mF and 22 mF. Each of these capacities has associated losses that influence the system efficiency.

$$E_{\text{stock}} = \frac{1}{2} C V^2$$

$$E_{\text{usable}} = \frac{1}{2} C (V_{\text{max}}^2 - V_{\text{min}}^2)$$

$$E_{\text{stock-no-usable}} = \frac{1}{2} C V_{\text{min}}^2$$

$$\text{So : } C = \frac{2E}{V_{\text{max}}^2} = 3,87\text{mF}$$

To store 44mJ, we have to choose a supercapacitor of 6,8mF, 10mF or 22mF.

For $E_{100\%} = 44\text{mJ}$:

With : $C = 6,8\text{mF}$

$$\text{We have : } E = \frac{1}{2} * 6,8 * 10^{-3} (4,3^2 - 2,2^2) = 44\text{mJ}$$

The minimum input DC power required :

$$P_{\text{IN min}} = 79 \mu\text{W}$$

For $E_{50\%} = 20\text{mJ}$:

With : $C = 2,2\text{mF}$

$$\text{We have : } E = \frac{1}{2} * 2,2 * 10^{-3} (4,8^2 - 2,2^2) = 20\text{mJ}$$

The minimum input DC power required :

$$P_{\text{IN min}} = 629 \mu\text{W}$$

For $E_{25\%} = 9,5\text{mJ}$:

With : $C = 1,5\text{mF}$

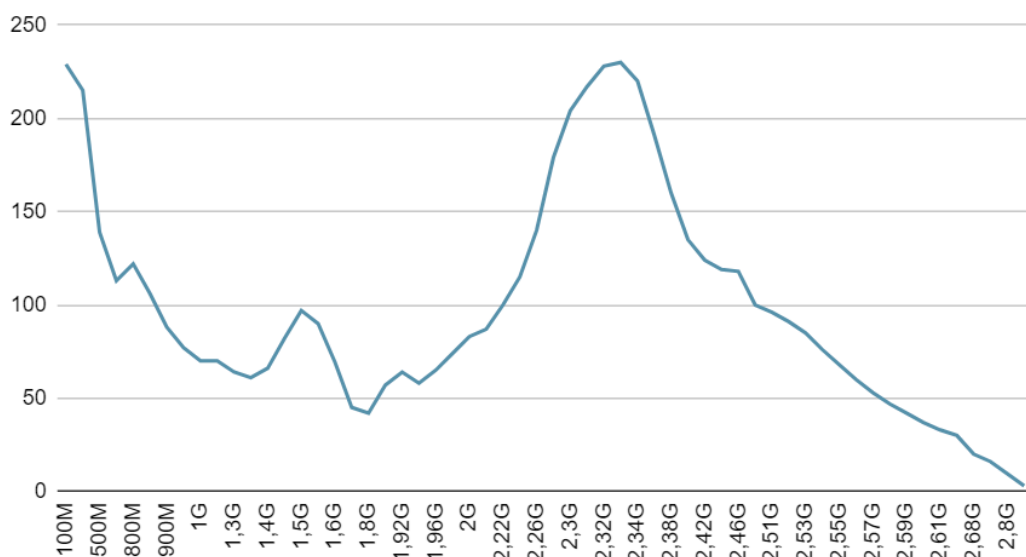
We have : $E = \frac{1}{2} * 1,5 * 10^{-3} (4,2^2 - 2,2^2) = 9,5\text{mJ}$

The minimum input DC power required :

$P_{\text{IN min}} = 432 \mu\text{W}$

II. Rectifier characterisation

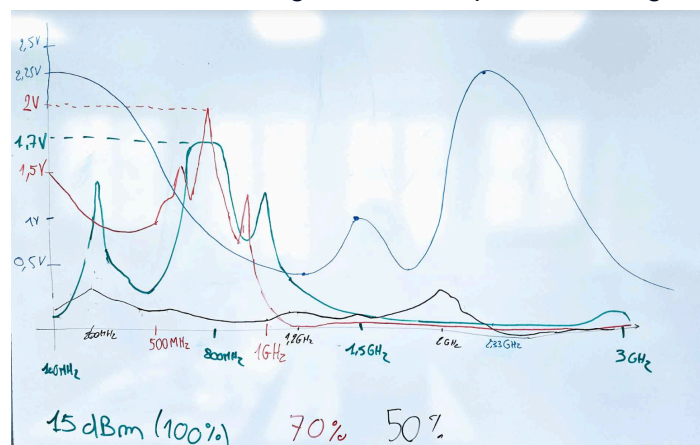
1. In this section, we first aimed to figure out if our rectifier performed better at 868 MHz or 2,45 GHz. By using a USRP device, we tested the rectifier at both frequencies and discovered it performs optimally at 2,45 GHz. Next, we varied the frequency between 800 MHz and 900 MHz, between 2.4 GHz and 2.5 GHz to measure how the voltage across the potentiometer changed with these adjustments.



Voltage at the ports of the potentiometer during a frequency sweep

We can conclude from the curve that the optimal frequency of our system is around 2,33 GHz. At this frequency, the LED achieves its brightest luminosity, indicating that the system is operating more efficiently.

If we now look at the curves of our colleagues and compare the charges:



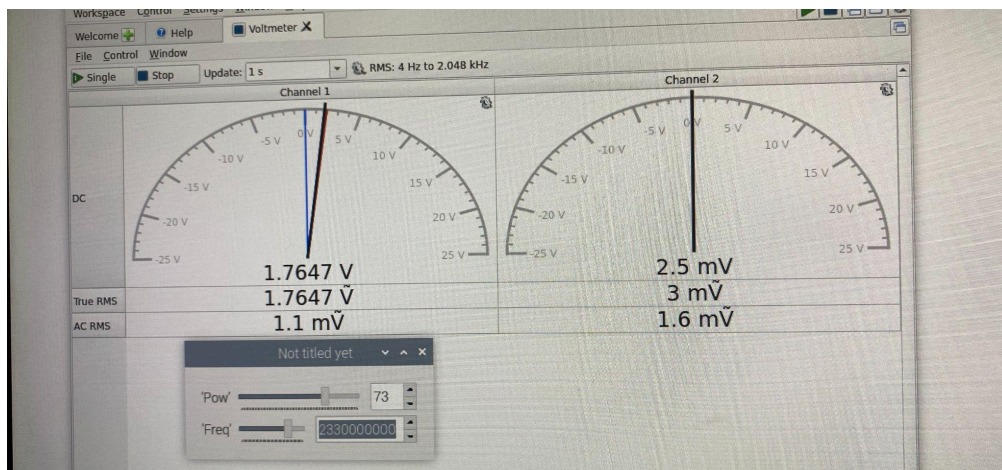
Red VS Green curve:

- Observation: The green curve corresponds to a configuration where only the LED is connected as a load, while the red curve represents a configuration where the PMU is in charge.
- Results:
 - Green curve: A voltage peak of 1.7V is observed at 800 MHz.
 - Red curve: The PMU system adapts better to the load, with a peak of 2V at 800 MHz.
- Interpretation:
 - The PMU allows for better voltage stabilization by optimizing the input, which explains the higher peak for the red curve.
 - While the green curve, with a single LED, shows less ability to reach high voltages, which is expected without advanced power management.

Black VS Blue curve:

- Observation: The black and blue curves have a similar overall shape, but a significant difference in output power and voltage.
- Results:
 - Blue curve (LED): The voltage peak reaches 2.25V at 2.3 GHz, which demonstrates better efficiency for this frequency.
 - Black curve (PMU): We observe a voltage peak of 0.25V at 2.3 GHz, much lower than that of the blue curve.
- Interpretation:
 - The blue curve, representing a configuration optimized for 2.45 GHz with a single LED, shows maximum efficiency for this frequency band, with a much higher peak voltage.
 - In contrast, the black curve, where the PMU system is associated with a supercapacitor, presents a much lower voltage at the same frequency, which shows that energy management via the PMU and the supercapacitor causes a drop in performance in this frequency band.

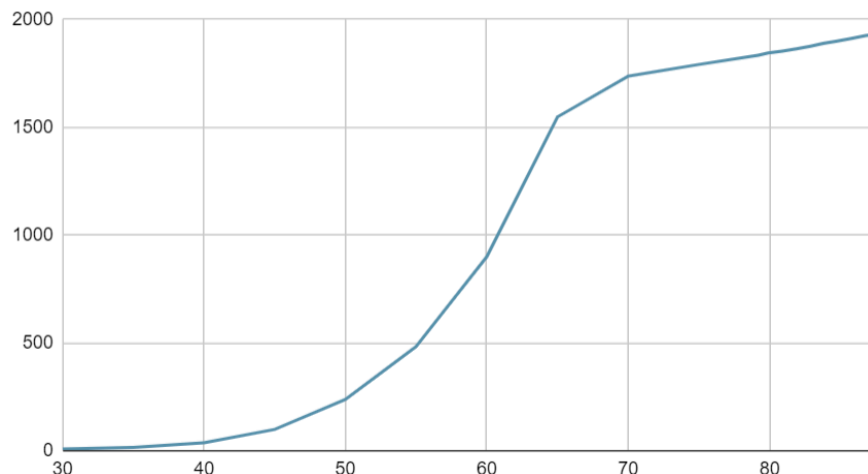
2. For our optimal frequency of 2,33 GHz and an RF input power of -15 dBm, we measured the voltage at the ports of the potentiometer during a load sweep : $U = 1,764 \text{ V}$.



Voltage for 2,33 GHz of frequency and -15dBm of RF input power

We measured also the voltage for : 2,45 GHz : $U = 1,769 \text{ V}$
868 MHz : $U = 1,746 \text{ V}$

3. We measured the voltage at the ports of the potentiometer during a RF power sweep (between -20 dBm and 0 dBm, with a step of 1 dBm) for the optimal frequency found previously and the optimal load :



Voltage at the ports of the potentiometer during a RF power sweep

We can analyze this curve in three parts:

- First phase $\Rightarrow$ Threshold voltage not reached: In this phase, the power supplied to the LED is insufficient to reach its threshold voltage. The voltage increases gradually, but the LED does not light up yet.
- Second phase $\Rightarrow$ Gradual saturation of the LED: When the power reaches the threshold value, the LED begins to light up. During this phase, the voltage continues to increase rapidly with the increase in power, and the brightness of the LED increases proportionally.
- Third phase $\Rightarrow$ Full saturation of the LED: Once the LED reaches its full brightness, the voltage stabilizes. Beyond this point, even if the power continues to increase, the voltage remains constant and the LED can no longer increase in brightness.

4. By using the targeted loads, the minimum RF input power allowing a good functioning is a gain of 67.

5. We didn't have the time to characterize the time required for a recharge versus the RF input power.

III. Antenna choice

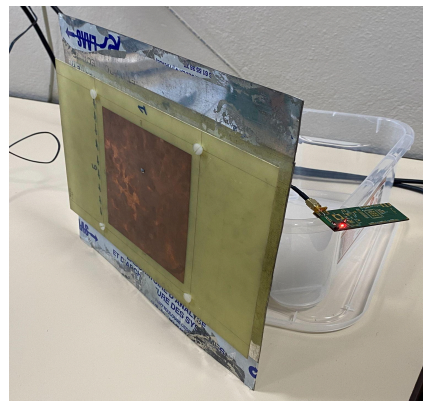
We had to choose the more appropriate antenna for our application. For this, we considered that most antennas are designed to be half the wavelength of the emitted frequency.

For the radiation pattern, directional antennas are ideal for fixed objects when the source direction is known, whereas omnidirectional antennas are better suited for scenarios where the object's position and source direction are unpredictable. If the source's polarization is known, the antenna's polarization should be aligned accordingly; otherwise, circular polarization is recommended, as it guarantees capturing at least half of the energy regardless of the source's polarization.

Through our tests, we observed that the little antenna had a good intensity for a distance of 3 meters and the large antenna had it for a distance of 5 meters. Also, the large antenna is ideal for flat surfaces (e.g., temperature sensors on walls, switches), but not suitable for non-flat surfaces (e.g., a mouse or similar objects).



Testing the little antenna



Testing the large antenna

The most suitable antenna for our use case, specifically to power a LED, is a directional antenna, as it effectively concentrates energy from a known source direction, maximizing the efficiency of energy harvesting over a fixed distance.

IV. Ambient electromagnetic energy harvesting

We didn't have time to do this part but we can deduce the test to be carried out:

1. By using a spectrum analyser (or an USRP) and for the chosen antenna, measure the power available at the input of the rectifier for the targeted frequency:

Using a spectrum analyzer with the selected directional antenna picking up at about 5 meters. The antenna is connected to the input of the rectifier and the measurement is carried out at the frequency that the antenna is optimized, either 868 MHz or 2.45 GHz. The values are then recorded according to the intensity of the signal received at a distance of 5 meters. If the captured power is high enough to power the LED, the measurement can then be compared with the system's power requirements.

V. Radiative electromagnetic wireless power transfer

1) We estimated the maximum reachable distance between a power source and the rectenna according to the targeted frequency (868 MHz or 2.45 GHz) thanks to the formulas given by the lab instructions :

Efficiency

$$\eta_{rectenna} = \frac{P_{DC_out}}{P_{RF_in}} = \frac{P_{DC_out}}{S \cdot A_{eff}}$$

Effective electric field

$$E = \frac{\sqrt{30 \cdot P_{TX} \cdot G_{TX}}}{d} (V \cdot m^{-1})$$

Incident electromagnetic power density

$$S = \frac{E^2}{120 \cdot \pi} (W \cdot m^{-2})$$

Effective area of the antenna

$$A_{eff} = G_{RX} \cdot \frac{\lambda^2}{4 \cdot \pi} (m^2)$$

$$d = \sqrt{\frac{P_{tx} \cdot G_{tx} \cdot G_{rx} \cdot c^2}{PRF_{in} \cdot 16 \cdot \pi^2 \cdot f^2}}$$

We found that the distance is : = 100m.

2) To power loads such as an LED, one can include a microstrip antenna matched to the frequency (868 MHz or 2.45 GHz) with an efficient rectifier circuit.

3) Optimize antenna gain: Use directional antennas or Adjust frequency: Test at different frequencies within the permitted range to maximize efficiency or Increase Power if it is allowed.

4) We Imagined some applications and deployment strategies of WSN powered by WPT.

For example, WSN can be used in hospitals to monitor patient vitals (like heart rate, blood pressure) or even in smart homes to track the well-being of elderly or chronically ill patients.

In healthcare facilities, WPT systems could provide power to sensors embedded in patient beds or wearable devices, ensuring that they remain operational without the need for regular recharging. WPT units could be embedded in ceilings or walls for seamless and continuous power delivery.

Conclusion

This lab focused on the feasibility of using wireless power transfer to power connected devices. We identified the optimal frequency for our rectifier, explored the power requirements of an LED, and assessed different antenna types for various applications. The findings highlight the potential for developing efficient, self-powered wireless sensor networks (WSNs). This approach can significantly enhance energy management and operational capabilities in diverse fields, including healthcare, smart homes, and industrial applications.